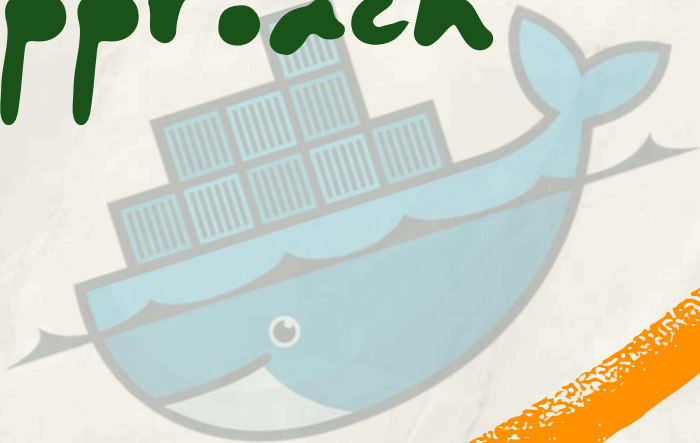




# BUILDING PRODUCTION READY KUBERNETES ON UBUNTU

A Checklist Approach



UbuCon India 2025  
Nov 15, 2026

Kunal Das

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# HALLO, I'M KUNAL

Developer Advocate APAC, CAST AI  
Organizer of  
CNCF Kolkata,  
Cloud Computing Circle,  
Hashicorp User Group Bangalore

7x Azure, 1x Hashicorp Certified, FinOps  
Certified Engineer





# PROBLEM

## TOO MANY CHOICES

- Which CNI? Calico? Cilium? Flannel?
- Storage: Local? OpenEBS? Longhorn?
- Security: Where do I even start?
- Monitoring: ELK? Prometheus? Both?

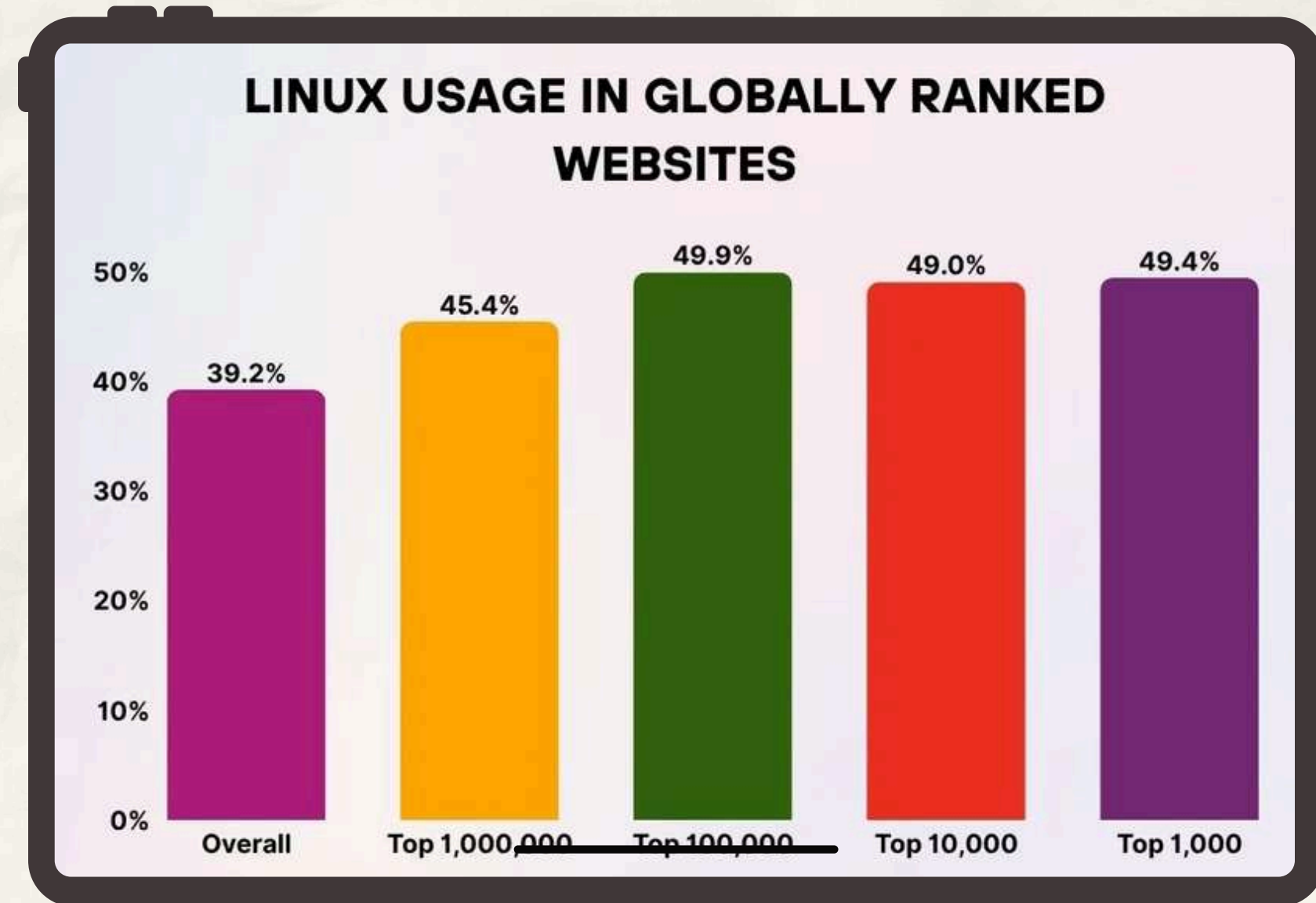
## WHAT GOES WRONG

- Late night production incidents
  - Security vulnerabilities
- No backups when you need them
- Mysterious pod crashes
- "It worked on my laptop!"

# WHY UBUNTU

## STATS

- AWS EC2 instances are **83.5%** Linux-based, with Amazon Linux, Ubuntu, and RHEL leading.
- Azure reports a **61.8%** Linux VM usage rate among active subscriptions in 2025.
- In Google Cloud, Linux VMs constitute **91.6%** of total virtual machine deployments.
- The Cloud Native Computing Foundation (CNCF) notes that **96.4%** of production Kubernetes clusters run on Linux.



# WHY UBUNTU



## STATS

How many years  
of enterprise  
support does  
Ubuntu Pro  
provide for an  
Ubuntu LTS  
release?



# WHY UBUNTU



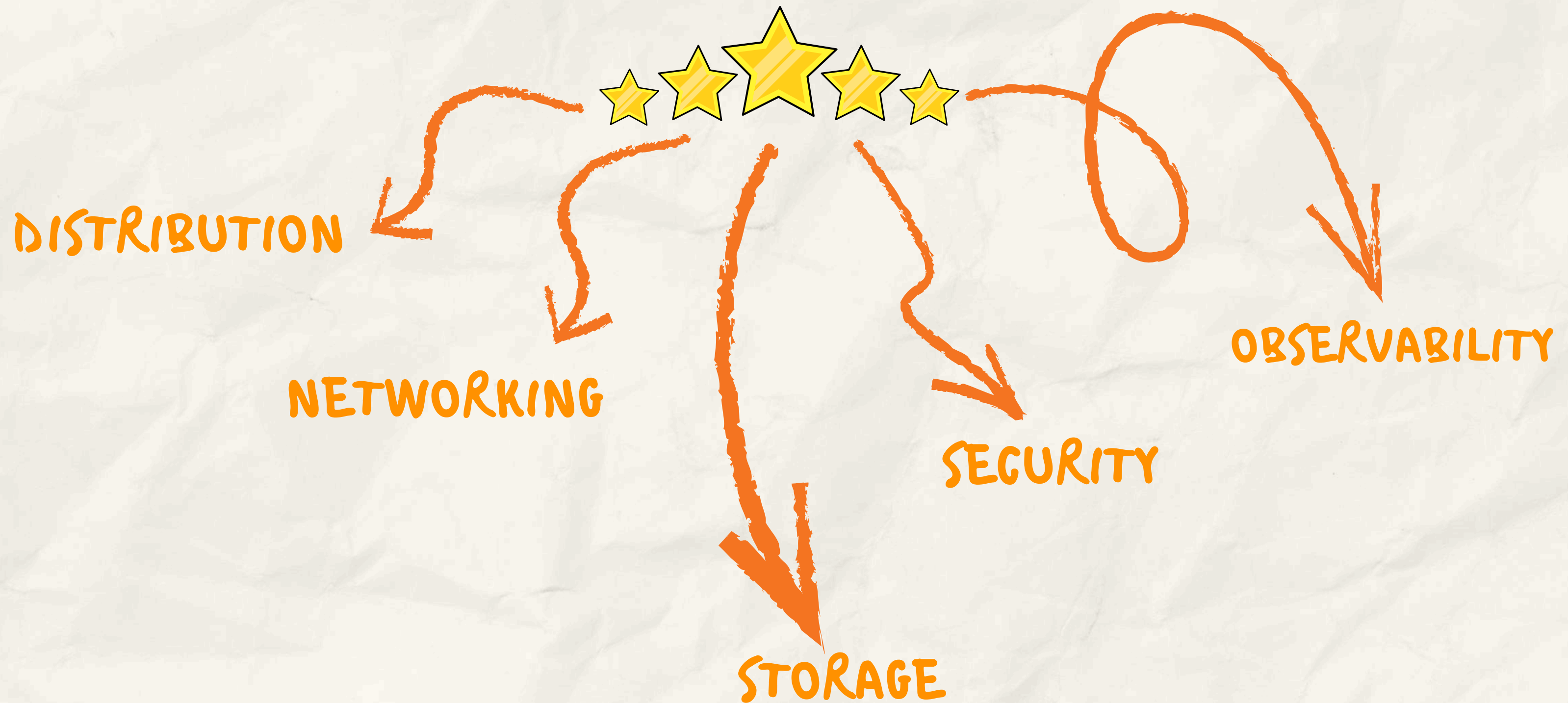
## BENEFITS

- 60% of K8s clusters run on Ubuntu
- Snap packages = Easy updates
- Fast mirrors in India.
- AppArmor built-in



```
ji@focal: ~  
ji@focal:~$ snap install vlc  
Download snap "vlc" (2103) from channel "stable" 11% 3.88MB/s 1m11s
```

# THE CHECKLIST FRAMEWORK

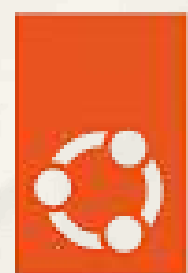




# THE CHECKLIST FRAMEWORK

## DISTRIBUTION

---



MicroK8s



Charmed Kubernetes



**SMALL/EDGE**  
**ONE COMMAND**  
**AUTO-UPDATES**

---



**DEV / SMALL**

**ENTERPRISE**  
**JUJU-MANAGED**  
**DAY-2 OPS**

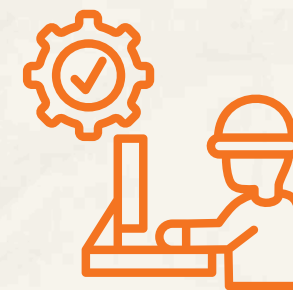
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**PRODUCTION**

**FULL CONTROL**  
**VANILLA K8S**  
**DIY EVERYTHING**

---



**LEARNING**

# THE CHECKLIST FRAMEWORK

## DISTRIBUTION

---



# THE CHECKLIST FRAMEWORK

## NETWORKING

---



cilium



flannel

NETWORK POLICIES  
MATURE + STABLE  
GREAT DOCS

---



SECURITY FIRST

EBPF-POWERED  
OBSERVABILITY++  
UBUNTU NATIVE

---



MODERN CHOICE

SIMPLE OVERLAY  
EASY SETUP  
LIGHTWEIGHT

---



GETTING STARTED



# THE CHECKLIST FRAMEWORK

## NETWORKING



```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: deny-all-ingress
spec:
  podSelector: {}
  policyTypes:
  - Ingress
```



# THE CHECKLIST FRAMEWORK

## STORAGE

---



FAST + LOCAL  
BUILT INTO  
MICROK8S  
ZERO CONFIG

---



DEV / TESTING



CLOUD-NATIVE  
EASY BACKUPS  
REPLICATED  
WEB UI

---



PRODUCTION



UBUNTU-BORN  
FLEXIBLE  
USES LVM/ZFS  
ENTERPRISE

---



SCALE + CONTROL



# THE CHECKLIST FRAMEWORK

## SECURITY

THE MOST IMPORTANT PILLAR

**RUNNING AS ROOT**

**NO RESOURCE LIMITS**

**DEFAULT PASSWORDS**

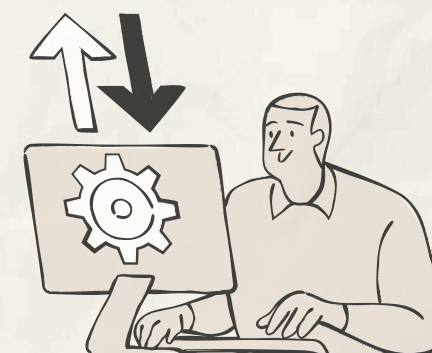
**OPEN TO INTERNET**

**NO BACKUPS**

**SKIPPING UPDATES**

Username  
Admin

Password  
123456I

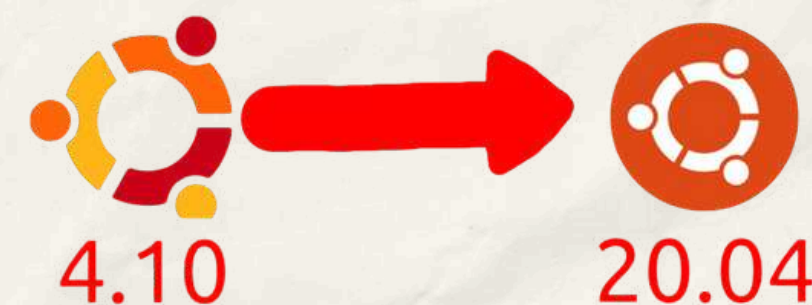




# THE CHECKLIST FRAMEWORK

## SECURITY

THE MOST IMPORTANT PILLAR



# THE CHECKLIST FRAMEWORK

## SECURITY

THE MOST IMPORTANT PILLAR

87% OF BREACHES FROM  
MISCONFIGURATION  
NOT HACKERS!



APP ARMOR



kubernetes  
Network Policies



## APP ARMOR

APPARMOR IS A LINUX SECURITY MODULE PRE-INSTALLED ON UBUNTU THAT ENHANCES SYSTEM SECURITY BY RESTRICTING APPLICATION CAPABILITIES THROUGH PER-PROGRAM PROFILES



### APPARMOR PROTECTS AGAINST:

- ✗ UNAUTHORIZED FILE ACCESS
- ✗ NETWORK CONNECTIONS
- ✗ PRIVILEGE ESCALATION
- ✗ RAW SOCKET ACCESS
- ✗ CAPABILITY ABUSE



### UBUNTU ADVANTAGE

- ✓ PRE-INSTALLED & ENABLED
- ✓ WORKS WITH DOCKER/CONTAINERD
- ✓ ZERO CONFIGURATION NEEDED
- ✓ PROFILES IN /ETC/APPARMOR.D/
- ✓ AA-STATUS SHOWS EVERYTHING





# APP ARMOR

## How AppArmor Works on Ubuntu

1



### Ubuntu Node

AppArmor enabled by default ✓



2



### Add Annotation

One line in Pod YAML



3



### Container Locked

Mandatory Access Control active



4



### Production Ready

Sleep well at night! 😴



# APP ARMOR

```
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: secured-nginx
5    annotations:
6      # ⚡ THIS LINE = SECURITY MAGIC!
7      container.apparmor.security.beta.kubernetes.io/nginx: localhost/k8s-restricted
8  spec:
9    containers:
10     - name: nginx
11       image: nginx:latest
```

That's the power of Ubuntu! One line = Enterprise Security ⚡

# THE CHECKLIST FRAMEWORK

## OBSERVABILITY



TIME-SERIES DB  
CPU/MEMORY/DISK  
ALERTS + RULES  
GRAFANA READY

---

WHAT'S  
HAPPENING?



**LOKI**

LIKE PROMETHEUS  
FOR LOGS  
10X CHEAPER!  
LABEL-BASED

---

WHY IT  
HAPPENED?



**JAEGER**

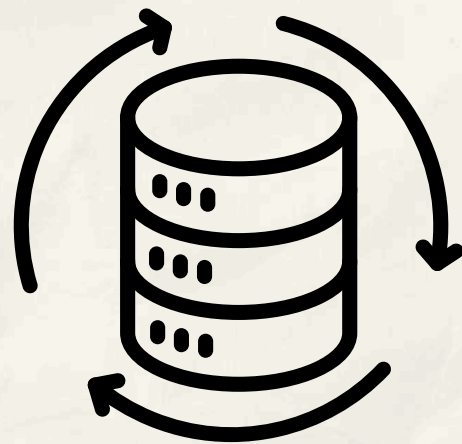
REQUEST FLOW  
MICROSERVICES  
FIND BOTTLENECKS  
DISTRIBUTED

---

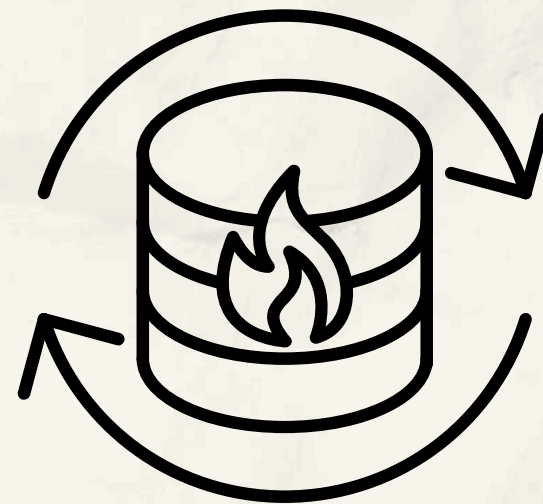
HOW IT FLOWS?



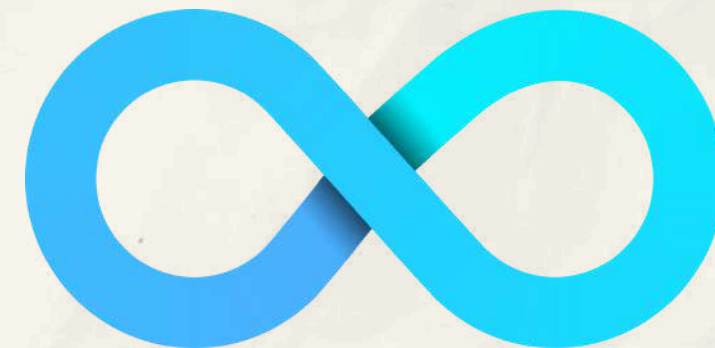
# A LOT REMAINS



UPDATES



DISASTER  
RECOVERY



CI-CD

```
microk8s enable loki
```

```
microk8s enable argocd
```

# THE COMPLETE UBUNTU K8S CHECKLIST

## ● UBUNTU FOUNDATION

- ✓ UBUNTU SERVER 22.04 LTS (OR 24.04 LTS)
- ✓ UBUNTU PRO ENABLED (FREE FOR PERSONAL USE)
- ✓ LIVEPATCH ACTIVATED FOR ZERO-DOWNTIME UPDATES
- ✓ SYSTEM HARDENING APPLIED (USG-HARDENING)

## ● KUBERNETES DISTRIBUTION

- ✓ MICROK8S FOR SMALL/EDGE OR CHARMED K8S FOR ENTERPRISE
- ✓ CNI ENABLED (CALICO/CILIUM)
- ✓ STORAGE ADDON CONFIGURED (HOSTPATH/OPENEBS)

## ● UBUNTU SECURITY

- ✓ APPARMOR PROFILES CREATED AND LOADED
- ✓ RBAC CONFIGURED WITH UBUNTU USER GROUPS
- ✓ NETWORK POLICIES IN PLACE
- ✓ POD SECURITY STANDARDS ENFORCED
- ✓ REGULAR APT UPDATE && APT UPGRADE
- ✓ VULNERABILITY SCANNING AUTOMATED



# THE COMPLETE UBUNTU K8S CHECKLIST

## ● OBSERVABILITY

- ✓ MICROK8S OBSERVABILITY ADDON OR PROMETHEUS STACK
- ✓ GRAFANA WITH UBUNTU DASHBOARDS
- ✓ LOKI FOR LOGS
- ✓ NODE-EXPORTER ON ALL UBUNTU NODES
- ✓ ALERTMANAGER CONFIGURED

## ● BACKUP + DR

- ✓ ETCD BACKUPS VIA UBUNTU CRON
- ✓ VELERO FOR APPLICATION BACKUPS
- ✓ TESTED RESTORE PROCEDURES
- ✓ OFF-SITE BACKUPS CONFIGURED

## ● OPERATIONS

- ✓ GITOPS WITH ARGOCD/FLUX
- ✓ CI/CD PIPELINE ON UBUNTU
- ✓ MONITORING + ALERTING TESTED
- ✓ RUNBOOKS DOCUMENTED





1. Try MicroK8s on your Ubuntu laptop this weekend
2. Join Ubuntu India community channels
3. Download the production checklist
4. Share your Ubuntu K8s stories at our next meetup!

**CALL TO ACTION**

THANK YOU  
FOR YOUR ATTENTION

PING ME  
ANYTIME TO  
KNOW MORE

